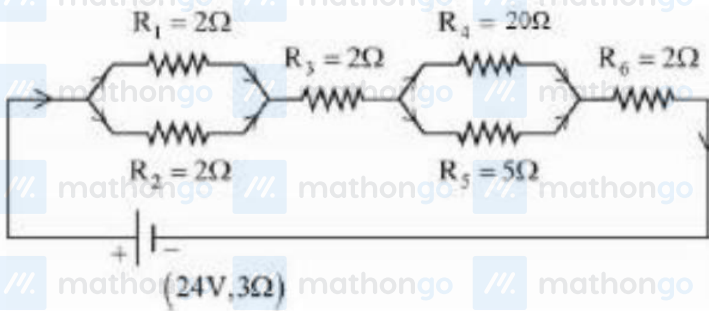


Q1 - 24 January - Shift 1

As shown in the figure, a network of resistors is connected to a battery of 24 V with an internal resistance of 3Ω . The currents through the resistors R_4 and R_5 are I_4 and I_5 respectively. The values of I_4 and I_5 are :



Space for your notes:

(1) $I_4 = \frac{8}{5} \text{ A}$ and $I_5 = \frac{2}{5} \text{ A}$

(2) $I_4 = \frac{24}{5} \text{ A}$ and $I_5 = \frac{6}{5} \text{ A}$

(3) $I_4 = \frac{6}{5} \text{ A}$ and $I_5 = \frac{24}{5} \text{ A}$

(4) $I_4 = \frac{2}{5} \text{ A}$ and $I_5 = \frac{8}{5} \text{ A}$

Q2 - 24 January - Shift 1

A hollow cylindrical conductor has length of 3.14 m, while its inner and outer diameters are 4 mm and 8 mm respectively. The resistance of the conductor is $n \times 10^{-3} \Omega$.

If the resistivity of the material is $2.4 \times 10^{-8} \Omega\text{m}$.

The value of n is .

Space for your notes:

Q3 - 24 January - Shift 2

A cell of emf 90 V is connected across series combination of two resistors each of 100Ω resistance. A voltmeter of resistance 400Ω is used to measure the potential difference across each resistor. The reading of the voltmeter will be :

- (1) 40 V
- (2) 45 V
- (3) 80 V
- (4) 90 V

Space for your notes:

Q4 - 24 January - Shift 2

If a copper wire is stretched to increase its length by 20%. The percentage increase in resistance of the wire is _____ %.

Space for your notes:

Q5 - 25 January - Shift 1

A uniform metallic wire carries a current 2 A. when 3.4 V battery is connected across it. The mass of uniform metallic wire is 8.92×10^{-3} kg. density is 8.92×10^3 kg/m³ and resistivity is $1.7 \times 10^{-8} \Omega\text{-m}$. The length of wire is :

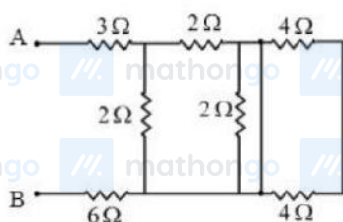
- (1) $l = 6.8$ m
- (2) $l = 10$ m
- (3) $l = 5$ m
- (4) $l = 100$ m

Space for your notes:

Q6 - 25 January - Shift 1

In the given circuit, the equivalent resistance between the terminal A and B is _____ Ω .

Space for your notes:



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Q7 - 25 January - Shift 2

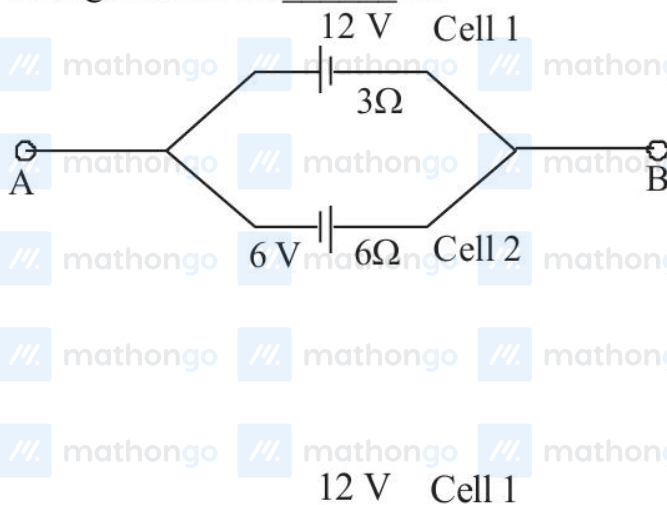
The resistance of a wire is $5\ \Omega$. Its new resistance in ohm if stretched to 5 times of its original length will be :

- (1) 625 (2) 5 (3) 125 (4) 25

Space for your notes:

Q8 - 25 January - Shift 2

Two cells are connected between points A and B as shown. Cell 1 has emf of 12 V and internal resistance of $3\ \Omega$. Cell 2 has emf of 6V and internal resistance of $6\ \Omega$. An external resistor R of $4\ \Omega$ is connected across A and B. The current flowing through R will be _____ A.



Space for your notes:

Q9 - 29 January - Shift 2

With the help of potentiometer, we can determine the value of emf of a given cell. The sensitivity of the potentiometer is

- (A) directly proportional to the length of the potentiometer wire
- (B) directly proportional to the potential gradient of the wire
- (C) inversely proportional to the potential gradient of the wire
- (D) inversely proportional to the length of the potentiometer wire

Choose the correct option for the above statements:

- (1) B and D only
- (2) A and C only
- (3) A only
- (4) C only

Space for your notes:

Q10 - 29 January - Shift 2

When two resistance R_1 and R_2 connected in series and introduced into the left gap of a meter bridge and a resistance of $10\ \Omega$ is introduced into the right gap, a null point is found at 60 cm from left side. When R_1 and R_2 are connected in parallel and introduced into the left gap, a resistance of $3\ \Omega$ is introduced into the right-gap to get null point at 40 cm from left end. The product of $R_1 R_2$ is _____ Ω^2

Space for your notes:

Q11 - 30 January - Shift 1

The charge flowing in a conductor changes with time as $Q(t) = \alpha t - \beta t^2 + \gamma t^3$. Where α, β and γ are constants. Minimum value of current is :

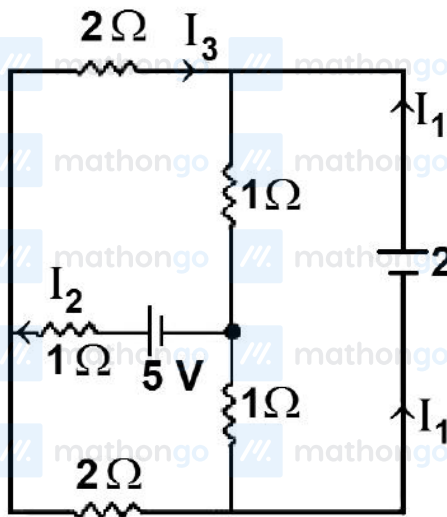
- (1) $\alpha - \frac{3\beta^2}{\gamma}$ (2) $\alpha - \frac{\gamma^2}{3\beta}$
(3) $\beta - \frac{\alpha^2}{3\gamma}$ (4) $\alpha - \frac{\beta^2}{3\gamma}$

Space for your notes:

Q12 - 30 January - Shift 1

In the following circuit, the magnitude of current I_1 , is _____ A.

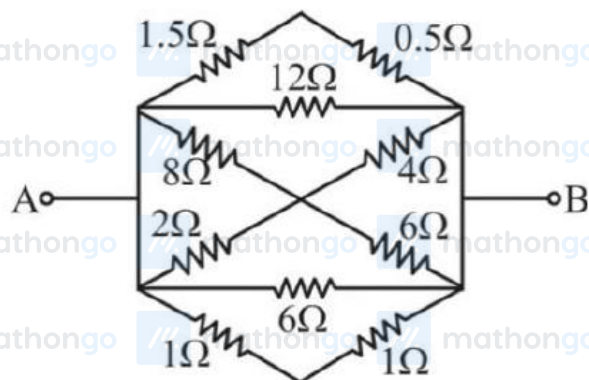
Space for your notes:



Q13 - 30 January - Shift 2

The equivalent resistance between A and B is

Space for your notes:



(1) $\frac{2}{3} \Omega$

(2) $\frac{1}{2} \Omega$

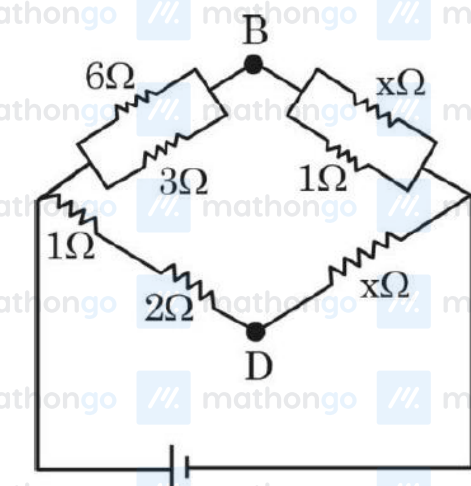
(3) $\frac{3}{2} \Omega$

(4) $\frac{1}{3} \Omega$

Q14 - 30 January - Shift 2

If the potential difference between B and D is zero, the value of x is $\frac{1}{n} \Omega$. The value of n is

Space for your notes:



x

Q15 - 31 January - Shift 1

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The drift velocity of electrons for a conductor connected in an electrical circuit is V_d . The conductor is now replaced by another conductor with same material and same length but double the area of cross section. The applied voltage remains same. The new drift velocity of electrons will be

- (1) V_d
- (2) $\frac{V_d}{2}$
- (3) $\frac{V_d}{4}$
- (4) $2V_d$

Space for your notes:

Q16 - 31 January - Shift 1

Two identical cells, when connected either in parallel or in series gives same current in an external resistance 5Ω . The internal resistance of each cell will be _____ Ω .

Space for your notes:

Q17 - 31 January - Shift 2

The amount of thermal energy is developed by a resistor in 10 s when a current of 4A is passed through it. If the current is increased to 16A, the thermal energy developed by the resistor in 10 s will be:

- (1) H
- (2) $16H$
- (3) $\frac{H}{4}$
- (4) $4H$

Space for your notes:

Q18 - 31 January - Shift 2

The number of turns of the coil of a moving coil galvanometer is increased in order to increase current sensitivity by 50%. The percentage change in voltage sensitivity of the galvanometer will be :

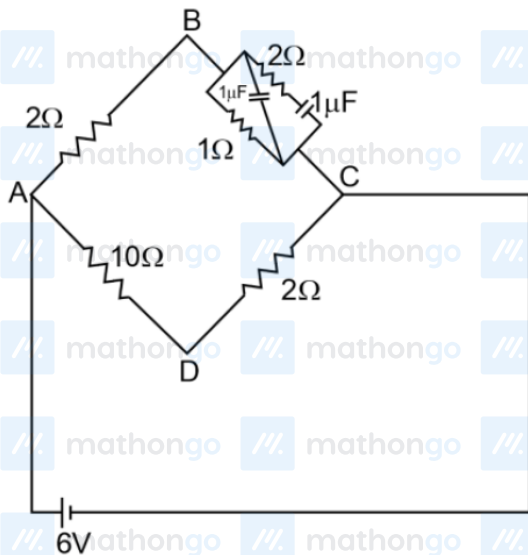
- (1) 100% (2) 50%
(3) 75% (4) 0%

Space for your notes:

Q19 - 31 January - Shift 2

For the given circuit, in the steady state,

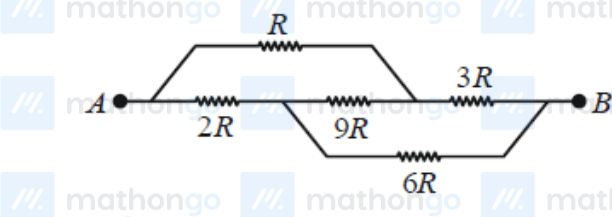
$$|V_B - V_D| = \underline{\hspace{2cm}} \text{ V.}$$



Space for your notes:

Q20 - 01 February - Shift 1

The equivalent resistance between A and B of the network shown in figure:



Space for your notes:

Q21 - 01 February - Shift 2

Given below are two statements : One is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A : For measuring the potential difference across a resistance of $600\ \Omega$, the voltmeter with resistance $1000\ \Omega$ will be preferred over voltmeter with resistance $4000\ \Omega$.

Reason R : Voltmeter with higher resistance will draw smaller current than voltmeter with lower resistance.

In the light of the above statements, choose the **most appropriate** answer from the options given below.

- (1) **A** is not correct but **R** is correct
- (2) Both **A** and **R** are correct and **R** is the correct explanation of **A**
- (3) Both **A** and **R** are correct but **R** is not the correct explanation of **A**
- (4) **A** is correct but **R** is not correct

Space for your notes:

Q22 - 01 February - Shift 2

Equivalent resistance between the adjacent corners of a regular n -sided polygon of uniform wire of resistance R would be:

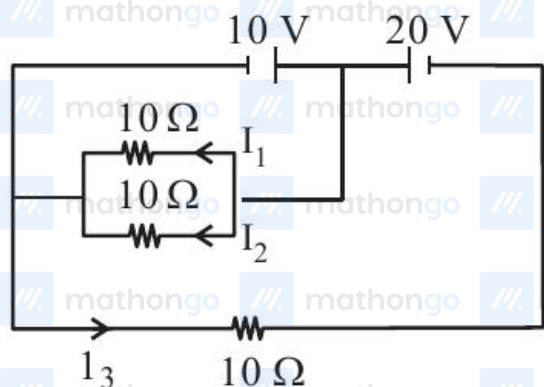
- (1) $\frac{(n-1)R}{n^2}$
- (2) $\frac{(n-1)R}{(2n-1)}$
- (3) $\frac{n^2R}{n-1}$
- (4) $\frac{(n-1)R}{n}$

Space for your notes:

Q23 - 01 February - Shift 2

In the given circuit the value of $\left| \frac{I_1 + I_3}{I_2} \right|$ is:

Space for your notes:



Answer Key

(As per Official NTA Key released on 2 Feb)

Q1 (4)

Q2 (2)

Q3 (1)

Q4 (44)

Q5 (2)

Q6 (10)

Q7 (3)

Q8 (1)

Q9 (2)

Q10 (30)

Q11 (4)

Q12 (1)

Q13 (1)

Q14 (2)

Q15 (1)

Q16 (5)

Q17 (2)

Q18 (4)

Q19 (1)

Q20 (4)

Q21 (1)

Q22 (1)

Q23 (2)

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Q1 (4)

Equivalent resistance of circuit

$$R_{eq} = 3 + 1 + 2 + 4 + 2$$

$$= 12\Omega$$

Current through battery $i = \frac{24}{12} = 2\text{A}$

$$I_4 = \frac{R_5}{R_4 + R_5} \times 2 = \frac{5}{20 + 5} \times 2 = \frac{2}{5}\text{A}$$

$$I_5 = 2 - \frac{2}{5} = \frac{8}{5}\text{A}$$

Q2 (2)

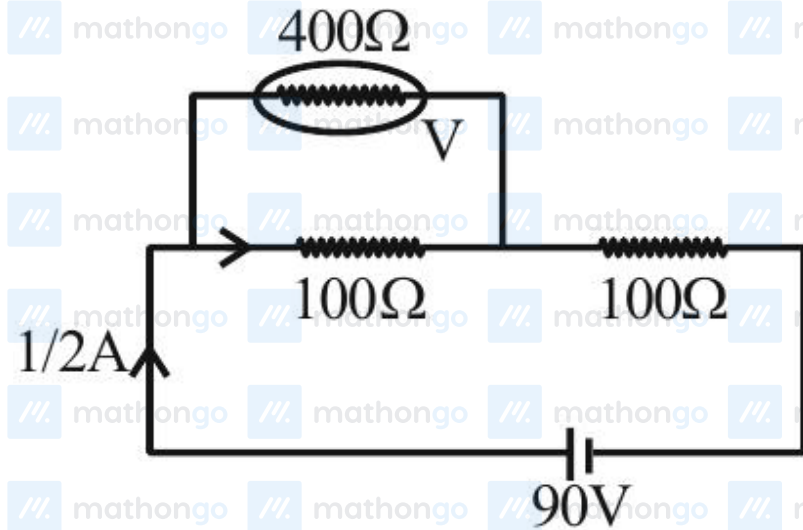
$R = \rho \frac{\ell}{A}$, the cross-sectional area is $\pi(b^2 - a^2)$

$$R = \rho \frac{\ell}{\pi(b^2 - a^2)} = \frac{2.4 \times 10^{-8} \times 3.14}{3.14 \times (4^2 - 2^2) \times 10^{-6}}$$

$$= 2 \times 10^{-3}\Omega$$

$$\rightarrow n = 2$$

Q3 (1)



$$R_{eq} = \frac{400 \times 100}{500} + 100$$

$$= 180\Omega$$

$$i = \frac{90}{180} = \frac{1}{2} \text{ A}$$

$$\text{Reading} = \frac{1}{2} \times \frac{400}{500} \times 100$$

$$= 40 \text{ volt}$$

Q4 (44)

As volume is constant,

So resistance $\propto (\text{length})^2$

$$\Rightarrow \% \text{ change in resistance} = 20 + 20 + \frac{400}{100} = 44\%$$

Q5 (2)

$$I = 2 \text{ A}$$

$$\Delta V = 3.4 \text{ V}$$

Using Ohm's Law

$$R = \frac{3.4}{2} = 1.7 \Omega$$

$$1.7 = \frac{\rho L}{A}$$

$$L = \frac{1.7(A)}{\rho}$$

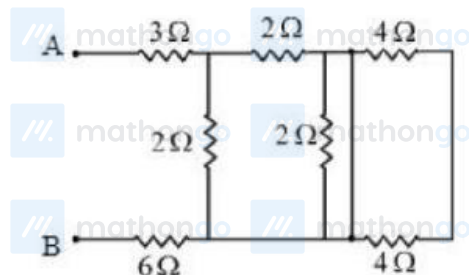
$$M = (\text{density volume})$$

$$\text{Volume} = \frac{8.92 \times 10^{-3}}{8.92 \times 10^3} = 10^{-6}$$

$$L^2 = \frac{1.7}{\rho} (10^{-6}) = \frac{1.7}{1.7} \times 10^2$$

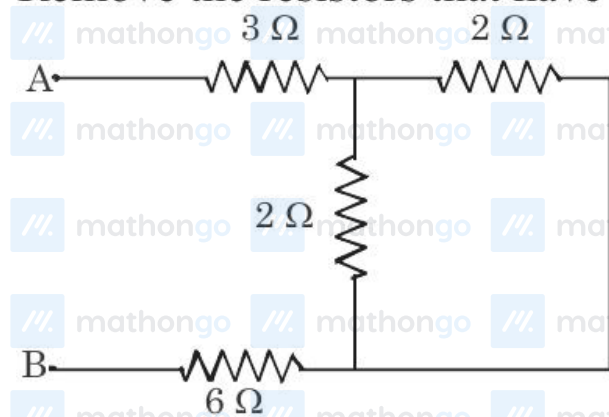
$$L = 10 \text{ m}$$

Q6 (10)



Both 4Ω resistance gets short.

Remove the resistors that have no current.



$$R_{eq} = 3 + (2 \parallel 2) + 6$$

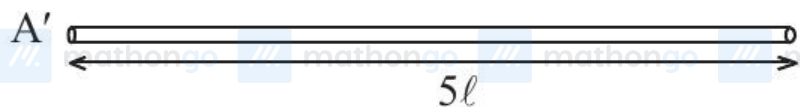
$$R_{eq} = 3 + 1 + 6$$

$$R_{eq} = 10\Omega$$

Q7 (3)



$$R_{\text{initial}} = \frac{\rho \ell}{A} = 5\Omega$$



$\therefore$ Volume of wire is constant in stretching

$$V_i = V_f$$

$$A_i \ell_i = A_f \ell_f$$

$$A\ell = A'(5\ell)$$

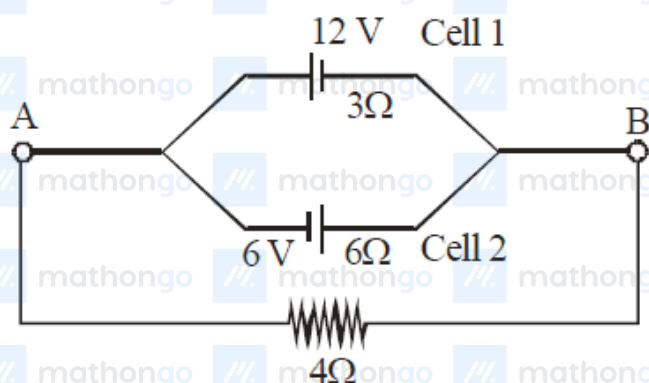
$$A' = \frac{A}{5}$$

$$R_f = \frac{\rho \ell_f}{A_f} = \frac{\rho(5\ell)}{\left(\frac{A}{5}\right)}$$

$$= 25 \left(\frac{\rho \ell}{A} \right)$$

$$= 25 \times 5 = 125 \Omega$$

Q8 (1)

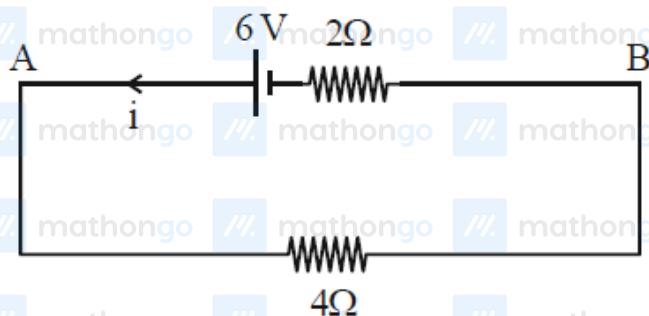


$$E_{eq} = \frac{\frac{12}{3} - \frac{6}{6}}{\frac{1}{3} + \frac{1}{6}}$$

$$E_{eq} = 6V$$

$$r_{eq} = 2\Omega$$

$$R = 4\Omega$$



$$\text{So, } i = \frac{6}{2+4} = 1\text{ A}$$

Q9 (2)

Sensitivity of potentiometer wire is inversely proportional to potential gradient.

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Q10 (30)

$$\frac{R_1 + R_2}{10} = \frac{60}{40} = \frac{3}{2} \Rightarrow R_1 + R_2 = 15$$

$$\text{Now } \frac{R_1 R_2}{(R_1 + R_2) \times 3} = \frac{40}{60} = \frac{2}{3} \Rightarrow R_1 R_2 = 30$$

Q11 (4)

$$Q = (\alpha t - \beta t^2 + \gamma t^3)$$

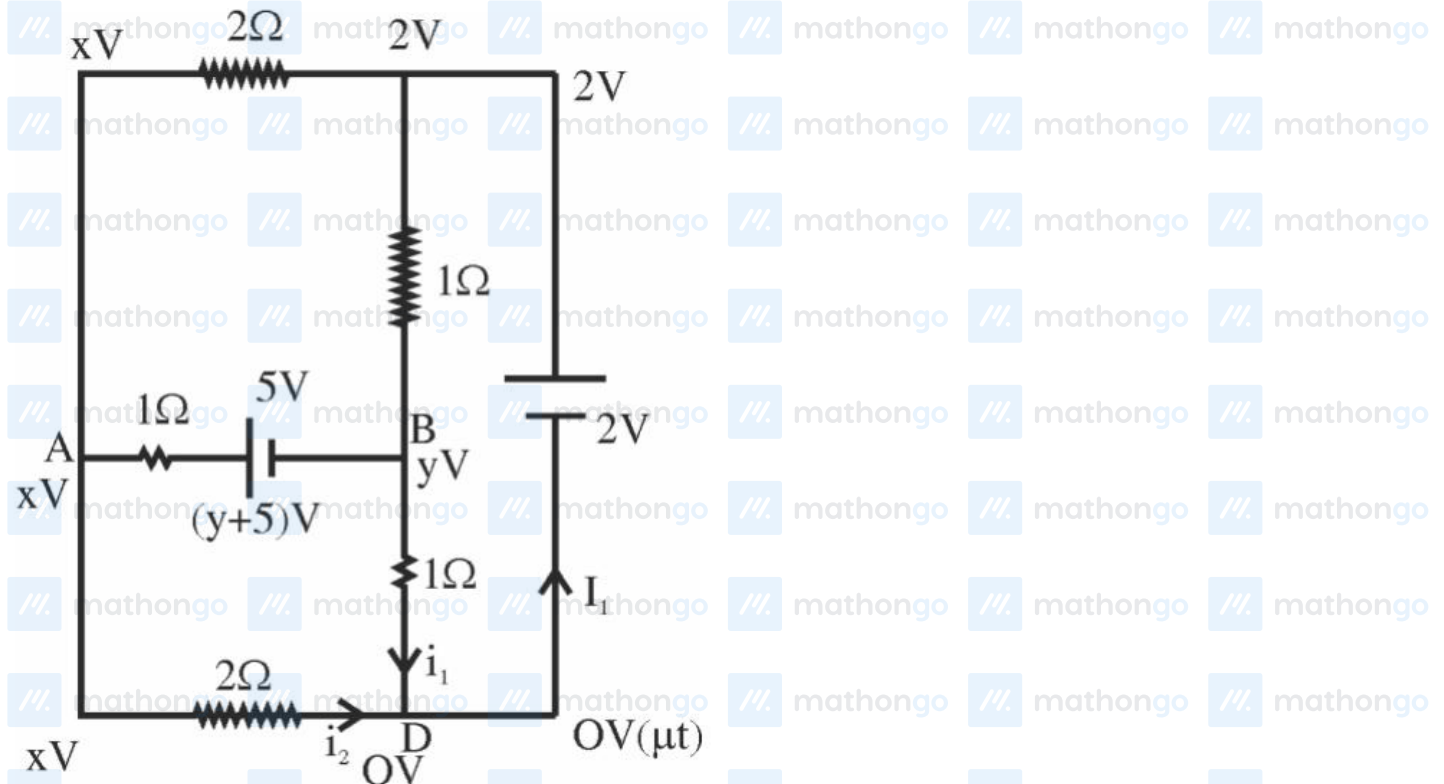
$$i = \frac{dQ}{dt} = (\alpha - 2\beta t + 3\gamma t^2)$$

$$\frac{di}{dt} = (3\gamma t - 2\beta) = 0$$

$$\Rightarrow t = \frac{\beta}{3\gamma}$$

$$i = (\alpha - 2\beta t + 3\gamma t^2) = \left(\alpha - \frac{\beta^2}{3\gamma} \right)$$

Q12 (1)



Junction law at A,

$$\frac{x - (y + 5)}{1} + \frac{x - 2}{2} + \frac{x - 0}{2} = 0 \quad \dots\dots(1)$$

Junction law at B,

$$\frac{y + 5 - x}{1} + \frac{y - 0}{1} + \frac{y - 2}{1} = 0 \quad \dots\dots(2)$$

On solving equation (1) and Equation (2)

$$x = 3$$

$$\& \ y = 0$$

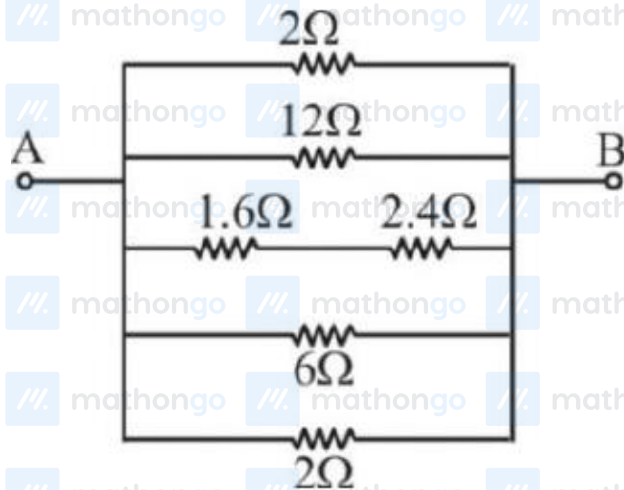
At D junction

$$I_1 = i_1 + i_2$$

$$I_1 = \frac{y - 0}{1} + \frac{x - 0}{2}$$

$$= \frac{0 - 0}{1} + \frac{3 - 0}{2}$$

Q13 (1)



$$\frac{1}{R_{eq}} = \frac{1}{2} + \frac{1}{12} + \frac{1}{4} + \frac{1}{6} + \frac{1}{2}$$

$$= \frac{6+1+3+2+6}{12} = \frac{18}{12} = \frac{3}{2}$$

$$\Rightarrow R_{eq} = \frac{2}{3} \Omega$$

Q14 (2)

$$\frac{2}{3} = \frac{x}{x+1}$$

$$\Rightarrow \frac{2}{3} = \frac{1}{x+1}$$

$$\Rightarrow x = 0.5 = \frac{1}{2}$$

$$n = 2$$

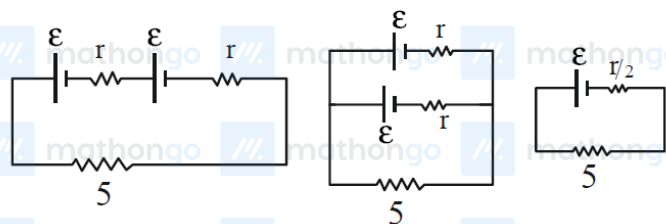
Q15 (1)

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$$V_d = \frac{eE}{m} \tau \text{ that is independent of area}$$

Q16 (5)

Parallel



$$i = \frac{2\varepsilon}{5 + 2r} \dots (1)$$

$$i = \frac{\varepsilon}{\frac{r}{2} + 5} \dots (2)$$

Equating (1) and (2)

$$\frac{2\varepsilon}{5 + 2r} = \frac{\varepsilon}{\frac{r}{2} + 5} \Rightarrow r + 10 = 5 + 2r$$

$$\boxed{r = 5} \quad \text{Ans. 5}$$

Q17 (2)

No Solution

Q18 (4)

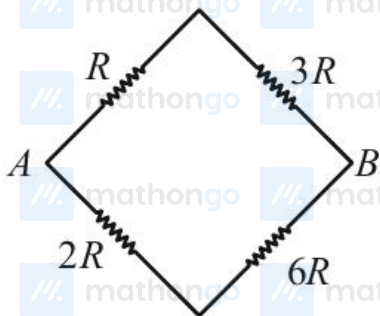
No Solution

Q19 (1)

No Solution

Q20 (4)

Wheat stone bridge is in balanced condition.



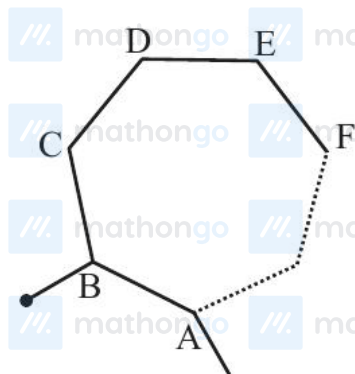
$$\frac{1}{R_{eq}} = \frac{1}{4R} + \frac{1}{8R}$$

$$R_{eq} = \frac{8R}{3}$$

Q21 (1)

Error of voltmeter decreases with increase in its resistance.

Q22 (1)



Suppose resistance of each arm is r , then $r = R/n$

$$R_{eq(AB)} = \frac{R_1 R_2}{R_1 + R_2}$$

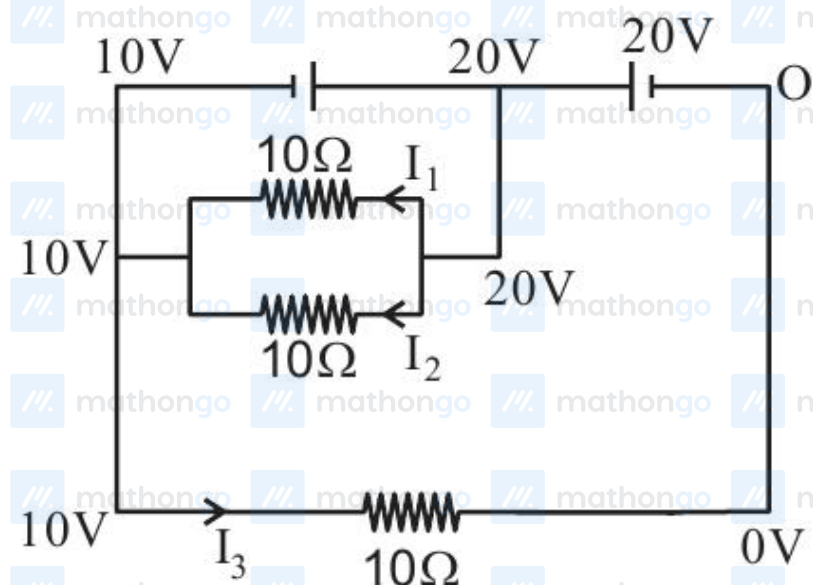
$$\frac{r(n-1)r}{r + (n-1)r}$$

$$= \frac{r(n-1)r}{nr}$$

$$= \frac{n-1}{n} r$$

$$= \frac{(n-1)R}{n^2}$$

Q23 (2)



$$I_1 = I_2 = \frac{20 - 10}{10} = 1\text{A}$$

$$I_3 = 1\text{A}$$

$$\left| \frac{I_1 + I_3}{I_2} \right| = 2$$